

Presentation 4

• **Links to presentation(s) and code(s) on GitHub:**

- [Code](#)
- [Julia's Dashboard](#)
- [Presentation Recording](#)
- [Presentation](#)

• **What did you do?**

This week, I verified Julia's work by walking through her logic step by step and confirming that each part of her code aligned with the intended methods. After validating her reasoning, I manually reproduced her calculations in python to ensure numerical accuracy, and found that her results were correct. Once her analysis was confirmed, I helped investigate several special-case questions by exploring the dataset directly in the spreadsheet, filtering and comparing values across relevant demographic and time-based segments, and examining patterns in call duration. This included looking at call by call breakdowns, activity and EP groups, and queue pathways to understand outliers or unexpected trends. Using sorting and contextual review of menu responses, I was able to identify and explain unusual patterns such as elevated call durations on specific days and missing menu values. Additionally I suggested we look at the non-zero queue times and when there is an agent name present for further downstream analysis.

• **How does it help the project?**

By verifying Julia's analysis manually by coding, calculations, and special-case insights, my work helped to strengthen the reliability of the overall analysis and ensured that any decisions

made from the data are rooted in correct logic. Confirming the hierarchical filters and call duration logic meant that the visualizations Julia produced and findings are aligned with the true caller experience. Together, this contributes to a more trustworthy foundation for optimizing call flows, refining menu design, and targeting improvements where they will have the most impact.

• Issues faced, Attempts to resolve issues, and Issues resolved

One of the main challenges faced was determining whether the logic and calculations used in the analysis were correct, especially given multiple layers of filtering and menu pathways. To address this, I manually retraced and re-coded Julia's calculations to verify numerical accuracy and confirm that her methodology matched the intended call-flow logic.

• Next steps

Building on Julia's progress, the next steps involve using the corrected hierarchical filter to explore trends beyond the seniors group and apply the same level of analysis to other menu pathways. This means breaking down call duration into menu, queue, and agent time for different caller segments and identifying whether similar patterns or inefficiencies appear elsewhere in the system.

• References (Mention if you built up on someone else's work)

- Julia Schaffner's SplitCallDuration code
- Code import files from Prof. Krishna
- Prof Krishna's queue logic

Presentation 3

• Links to presentation(s) and code(s) on GitHub:

- [Code](#)
- [Dashboard](#)
- [Presentation](#)

• What did you do?

In this code, I used the CAR dataset to identify and analyze callback events, specifically if they were picked up or not (i.e. if the customer was never reached in a callback). I started by converting the activity timestamps into proper datetime format, then filtered the data to include only contact sessions that contained a callback event. I made sure to keep all rows that were associated with that contact session ID, even if previous menu options were explored before. For each session, I removed all records that occurred at or after the first callback timestamp and created a unique callback instance by combining the session ID and date. This ensured that individual callbacks were being counted based on date and not on contact session ID. Next, I flagged whether each callback instance was “picked up”, meaning it included a LegalServerScreenPop activity and had an associated agent, and whether it involved a CallbackRetry. I combined these flags into a summary table and calculated overall metrics like total callbacks, pickup and retry counts, and their respective rates. Finally, I merged the flags back into the detailed dataset and exported both a detailed version and a summarized version (one row per callback instance) for visualization in Power BI.

To create my pie chart visualization in Power BI, I used the summary dataset I created that contained one row per callback instance, including whether each call was picked up or not. I imported this dataset into Power BI and inserted a pie chart visual. I set “Call Picked Up” as the legend and the count of callback instances as the values. This allowed me to compare the number of callbacks that were successfully picked up versus those that were not. I then formatted the chart to display both the counts and percentages, added labels for clarity, and included a large text card showing the percentage of callbacks not picked up (26.33%) for quick interpretation.

- **How does it help the project?**

My analysis helps Legal Aid by identifying a key area of improvement in the callback process. More specifically, the high percentage of callbacks that are not being picked up (26.33%). By quantifying how often callbacks fail to connect, I was able to highlight inefficiencies that lead to wasted agent time and poor customer experience. These findings directly support the recommendation to introduce customer notification and confirmation options, such as allowing customers to confirm or reschedule their callback time through an automated phone prompt. Implementing this feature would help ensure callbacks occur when the customer is actually ready to answer, reducing missed connections and improving overall callback success rates.

- **Issues faced, Attempts to resolve issues, and Issues resolved**

At first, I calculated the amount of pickups based on contact session ID, however I realized that treating the calls as separate calls based on different days would make more sense. Additionally, I saw that LegalScreenServerPop had popped up in different EP names other than Courtesy Callback Telephony EP meaning that I was counting the calls that might’ve been picked up but

also had a callback. I fixed this by deleting the rows before the first instance of Courtesy Callback Telephony EP.

- **Next steps**

My next steps would be to investigate the 26% of callbacks that were not picked up. I want to look into the demographics of the people that are not being reached when they are called back. By creating a hierarchical filter for menu options, I can see what the customer originally called for. Additionally I can investigate how many times the contact session ID was called back before they stopped trying to reach them, and then calculate the average. Additionally I can look into what proportion of callbacks were being missed at what time of day, and what times callbacks are being picked up. If there is a higher chance that people are picking up their callbacks at a certain time of day, my recommendation could be more optimized.

- **References (Mention if you built up on someone else's work)**

Used the CAR_data_import_code file to combine the datasets, created by Prof. Krishna.

Included the new September 2025 data.

Presentation 2

- **Links to presentation(s) and code(s) on GitHub:**

- [Presentation 2](#)
- [Dashboard](#)

- [Code](#)

• What did you do?

I first re-ran the CAR import data code to include the new call data from the month of September. Using that dataset in python, I ensured that the activity timestamps were in a proper datetime format to allow accurate time-based analysis. Then, I grouped the data by each contact session to calculate key call metrics: the earliest timestamp of the session (Call_Start_Time), the starting hour of the call, the number of unique activity timestamps within the session, and the total call duration in minutes. Finally, I created separate columns for the year, month, and day of the week for each call, enabling me to create a hierarchical filter in PowerBI.

I then grouped the call data by each contact session to summarize key indicators, including whether a session had a CallbackRetry, a LegalServerScreenPop, or an assigned agent. I then applied a logical rule to infer whether the call was likely picked up: sessions with a CallbackRetry and no agent were marked as not picked up, while sessions with either a LegalServerScreenPop or an agent were marked as picked up. Afterward, I checked for contradictions where the observed activity patterns did not align with this assumption, such as sessions containing both CallbackRetry and LegalServerScreenPop or CallbackRetry alongside an agent, and compiled those cases for inspection, ultimately printing the total sessions, the number of contradictions found, and a preview of those contradictory sessions.

In PowerBI, I put the Starting Hour on the x-axis filtering for times from 8-17, and Count of EP Name, filtering by Courtesy Callback Telephony EP. I added the Picked Up column as the legend. I then created a slicer to include year, month, then day of the week.

• How does it help the project?

My analysis helps Legal Aid by identifying patterns in callback timing and pickup rates, revealing that the current scheduling process is misaligned with customer availability. The analysis reveals that a large share of callbacks occur early in the morning when pickup rates are low, and that unanswered callbacks exceed completed ones across all hours. These insights provide a data-driven basis for improving the callback process through better scheduling, smarter retry logic, and potentially implementing confirmation or rescheduling options.

- **Issues faced, Attempts to resolve issues, and Issues resolved**

Determining if a callback was picked up → resolved by determining that CallbackRetry and LegalServerScreenPop in combination with presence of agent name could serve as an indication of callback pickup. Need to clarify this with Cynthia however.

- **Next steps**

My next steps will be to analyze agent-level performance with the objective of determining if certain agents have higher callback pickup rates and uncovering patterns in missed callbacks. To achieve this, I will visualize pickup performance by agent by building a stacked bar chart in Power BI that displays the number of callbacks picked up versus those missed for each agent. This approach will allow me to identify performance trends, highlight agents who consistently achieve higher pickup rates, and pinpoint areas where additional support or training may be beneficial.

- **References (Mention if you built up on someone else's work)**

Used the CAR_data_import_code file to combine the datasets, created by Prof. Krishna.

Included the new September 2025 data.

Presentation 1

• Links to presentation(s) and code(s) on GitHub:

- [Google Slides Presentation 1](#)
 - See Julia Schaffner's documentation for a link to the ppt presentation with PowerBI embedded visualizations. The google slides only contain the static graphics.
- [Dashboard 1](#)
 - All coding was done in Power BI

• What did you do?

I collaborated with Julia in order to create the visualization “No Answer From Customer” count versus “Starting Hour” bar graph. I first started by utilizing the code provided by Prof. Krishna to compile the All Calls Data by month into one single csv file. I also ran the code once again provided by Prof. Krishna to compile the CAR flow into a singular csv file. I then uploaded the data to PowerBI, but ultimately chose to work within Julia's PowerBI dashboard for efficiency and timing reasons.

CAR Data:

To prepare the data for further analysis in Power BI, Julia first used the “Group By” function to ensure that each *Contact Session ID* corresponded to a single row, preventing double counting. She then applied the “Advanced Group By” function to perform multiple aggregations, creating four new columns: *Count*, *Call Start Time*, *Starting Hour*, and *All Rows*. Finally, Julia

used the “Expand All Rows” feature to access and visualize the nested information within the dataset.

I then plotted the *Starting Hour* against the count of *EP name* filtering to only the *Courtesy Callback Telephony EP* to visualize call activity patterns of callbacks throughout the day. I filtered the data to include only working hours, from 8:30 a.m. to 5:00 p.m., using solely the CAR data source for this analysis. To refine the results, I applied the *Termination Reason* field as a legend, selecting only cases labeled “agent left” and “customer left” This allowed me to focus specifically on unsuccessful callbacks (where either the agent or customer left the call) within regular business hours.

I additionally plotted *Starting Hour* against the count of *Agent Name* to visualize the number of agents available throughout the day. I once again filtered the data to include only working hours, from 8:00 a.m. to 5:00 p.m. I did not include the rows that *Agent Name* was blank.

• **How does it help the project?**

This analysis helps the project by revealing that customers are leaving at the same rate agents are leaving despite having a relatively consistent amount of agents each hour of the day. When the callback volume is the highest in the first hour, we see the majority of customers leaving the call which makes sense. However when the call volume decreased, customers were leaving the calls at a proportional rate to the first hour of the day. I compared this to the number of agents staffed at each hour of the day. While most drop-offs occur during the first hour of high callback volume, the continued leave rates later cannot be explained by the slight decrease in agents. This suggests that factors like agent responsiveness, rather than staffing levels alone, are driving customer drop-offs and should be the focus of improvements.

• **Issues faced, Attempts to resolve issues, and Issues resolved**

The total number of callbacks was 72,426, of which 44 were terminated because the customer did not answer. Since termination reasons are only recorded for calls made after March 16, 2025, the analysis focuses on this subset of 11,916 calls. This timeframe is a drawback, only allowing me to explore 17% of the data, however it is also beneficial because it provides clearer insight into why certain callbacks are not being completed. However, with only 44 instances of customers not answering, this group is too small to generate actionable recommendations hence why I switched from my original recommendation. Therefore, the analysis was expanded to include termination reasons that represent larger proportions of the overall dataset such as “agent left” and “customer left”.

• **Next steps**

- Segment by call type or customer profile
 - Identify whether certain types of calls or customer groups are more likely to leave, which could reveal certain patterns
- Refine dashboard using other filters to find if other EP names follow a similar hourly pattern

• **References (Mention if you built up on someone else's work)**

Used the CAR_data_import_code file and AllCallsData_import_code to combine the datasets, created by Prof. Krishna.